

Technical Exploration and Gap Analysis for Generatively Designed Modular Autonomous Mobile Robots (SIH26112)

Problem Restatement and Technical Brief Analysis

The Smart India Hackathon (SIH26112) problem statement, sponsored by Autodesk within the hardware track, requires participating engineering teams to research, conceptualize, and structurally optimize a next-generation Autonomous Mobile Robot (AMR) / Automated Guided Vehicle (AGV) platform tailored for smart warehouse logistics¹. The brief mandates a fundamental departure from conventional subtractive manufacturing paradigms, challenging designers to re-imagine mobile robotic architectures by integrating generative design, topology optimization, multi-objective simulation, and Design for Additive Manufacturing (DfAM) within Autodesk Fusion¹.

The technical scope demands the simultaneous development of two interdependent, mandatory concept development challenges¹:

Part A focuses on the Universal Mobile Robot Platform (AGV/AMR Base)¹. This task entails designing a universal, load-bearing mobile base chassis intended to serve as a standardized, multi-functional mobility platform across diverse warehouse workflows¹. The base chassis must be engineered to support a variety of swap-capable top modules while achieving a balance between structural rigidity, carrying capacity, self-weight reduction, mechanical modularity, additive manufacturability, and maintenance accessibility¹.

Part B focuses on the Modular Functional Attachment¹. This task requires the concept design of at least one specialized top-side functional attachment that docks directly with the universal base platform¹. Exemplary operational modalities include automated bin handling, pallet lifting, motorized conveyor rollways, multi-axis robotic arm picking, optical inventory scanning, or site inspection modules¹.

The design brief establishes six core technical constraints that govern the unified system assembly¹. Structural integrity requires that the base chassis and top interface withstand high static payload forces, dynamic braking shear, cornering torsional moments, and vibrational impacts from uneven warehouse flooring¹. Payload capacity mandates maximizing the ratio of carried load to total structural self-weight¹. Weight reduction demands the systematic elimination of non-load-bearing structural mass using algorithmic material distribution to optimize wheel motor torque, battery efficiency, and acceleration profiles¹. Modularity dictates that the mechanical, electrical, and control interconnect between Part A and Part B must support repeatable mounting and unmounting without requiring structural modification of the base chassis¹. Manufacturability requires strict adherence to DfAM principles, accounting for self-supporting overhang angles, print bed orientation, anisotropic Z-axis material strength,

support structure removal, and post-processing access¹. Ease of maintenance mandates that internal sub-assemblies—including drive motors, battery enclosures, and central electronics—remain accessible for field inspection or rapid replacement without compromising frame stiffness¹.

Evaluation Rubric Category	Allocated Marks	Strategic Implication for Computational Engineering
Use of Advanced Fusion Features (Generative Design, Topology Optimization, Additive Build, Simulation)	30	Primary evaluation weight; demands multi-load-case algorithmic optimization rather than manual CAD modeling ¹ .
Design Optimization for 3D Printing (DfAM)	20	Requires additive-native geometry, such as structural lattice infills, organic force-conduits, and support-free overhangs ¹ .
Design Complexity and Workability	20	Mandates realistic dynamic kinematics, physical load path viability, and mechanical feasibility under load ¹ .
Innovative Design Features	15	Requires novel quick-change mechanical interfaces, integrated latching, or multifunctional structural features ¹ .
Selection of AM Process & Materials	15	Requires material matching (e.g., CF-PA12, AlSi10Mg) aligned with specific additive process physics ¹ .

An examination of the problem statement highlights several underspecified design parameters that require explicit engineering assumptions prior to simulation setup¹. First, the brief does not define a standard payload mass class, leaving the target capacity open across light-duty tote carriers (50 kg), medium logistics platforms (250 kg), and heavy pallet movers (1000+ kg)¹.

Defining these static mass boundaries and dynamic load vectors (acceleration shear, cornering torque, dynamic stopping) is necessary to construct constraint bounds in Autodesk Fusion³. Second, a operational discrepancy exists between full-scale industrial additive manufacturing (such as Selective Laser Sintering or Powder Bed Fusion) and the Grand Finale requirement to present scaled-down physical 3D prints¹. The conceptual design must be optimized for full-scale material properties while retaining structural geometric fidelity when scaled down for desktop FDM/SLA printing¹. Third, the brief leaves the electrical power interface, digital control bus pass-throughs, and safety interlocks unstandardized, necessitating the establishment of an integrated mechanical-electrical interconnect standard¹.

Existing Landscape Mapping

To contextualize the state of the art, existing solutions are analyzed across three key tiers: commercial AMR/AGV platforms, academic/research quick-change prototypes, and computational DfAM/generative design implementations in mobile robotics.

Commercial AMR architectures are dominated by heavy industrial platforms designed primarily using conventional subtractive sheet-metal and plate-weldment techniques². Mobile Industrial

Robots (MiR) produces the MiR250 platform, which features an **800 × 580 × 300 mm** footprint, a self-weight of approximately **78 kg** to **94 kg**, and a maximum payload capacity of **250 kg**². The MiR250 mechanical interface relies on a flat steel top plate featuring a standardized grid of blind threaded holes². Top modules, such as the ROEQ TMC500 or TR500 Auto lifters, are mounted via semi-permanent manual bolting, while electrical power, Safety PLC STO signals, and Ethernet communications pass through discrete side-port connectors².

Similarly, OTTO Motors produces the OTTO 100 (**150 kg** payload) and OTTO 1500 (**1900 kg** payload) platforms⁷. These systems utilize heavy-gauge structural steel frames paired with integrated lifting pins or top-mounted bolt patterns⁷. While commercially mature (TRL 9), these platforms exhibit low structural mass efficiency, high self-weight, and require manual tool-bound operations to swap functional top modules².

Academic and laboratory research focuses primarily on rapid, tool-less mechanical coupling and self-reconfigurable robotic joints¹⁴. Precision robotic tool changers (ATCs) and kinematic couplings utilize exact constraint principles, pairing three V-grooved surfaces with spherical elements to constrain six degrees of freedom with sub-micron positional repeatability¹⁵.

Pre-load force is typically maintained through high-pressure pneumatic draw-bars, permanent magnet arrays, or motorized ball-locking cams¹⁴. Modular mobile prototypes, such as the EMERGE system, demonstrate autonomous module docking using active mechanical grippers and magnetic latches¹⁴. However, these research concepts (TRL 3–4) are engineered for small manipulator arms or lightweight mobile units (<5 kg payload)¹⁴. Their reliance on localized point or line contacts generates high Hertzian contact stresses, rendering them unsuited for direct

fabrication in additive polymers without heavy metallic inserts⁴.

In the domain of computational design and additive manufacturing, generative design algorithms and topology optimization have been applied almost exclusively to isolated sub-components rather than complete vehicle systems¹⁷. Studies utilizing Autodesk Fusion Generative Design demonstrate structural mass reductions between 15% and 40% for robotic manipulator arms, drone frames, and wheel suspension uprights while preserving structural stiffness under defined static load cases¹⁷. These components are produced using Industrial Additive Manufacturing technologies, including Direct Metal Laser Sintering (DMLS) in AlSi10Mg aluminum alloy or Selective Laser Sintering (SLS) in carbon-fiber-reinforced polyamides (CF-PA12)⁴. However, these applications operate as isolated structural brackets bolted onto conventional frames, lacking system-level chassis integration, integrated cable conduits, or functional quick-docking mechanisms³.

Solution Name / Source	System Type	Attachment / Docking Mechanism	Manufacturing & Material Method	Key Structural / Operational Limitation
MIR250 / ROEQ System [cite: 2, 9, 10]	Commercial Base + Top Module	Semi-permanent bolt pattern grid; discrete multi-pin connectors for 24V/48V power, Safety PLC, and Ethernet ² .	Structural sheet-metal steel bending and frame welding; CNC plate milling; vacuum-formed covers ² .	High structural self-weight (~78-94 kg base); requires manual hand tools for module changes; non-optimized flat-plate load transfer ² .
OTTO 100 / OTTO 1500 [cite: 7, 8, 12]	Commercial AMR Base	Frame-integrated bolt grid; central lifting pins; ruggedized industrial cable harnesses ⁷ .	Heavy-gauge structural steel plate welding; CNC cross-member machining ⁷ .	Significant dead-weight penalty; complete absence of DfAM or generative optimization; tool-bound top module swaps ⁷ .

Comau Agile 1500 [cite: 8]	Commercial AGV Base	Standardized mechanical interface plates for towing, lifting tables, or roller conveyors ⁸ .	Structural steel box-frame fabrication and heavy axle assembly ⁸ .	Rigid non-optimized frame geometry; heavy mass; requires manual structural modifications for functional re-tasking ⁸ .
Kinematic Couplings & ATCs [cite: 15, 16]	Academic / Industrial Tool Interface	Three-groove exact constraint coupling or ball-locking draw-bar driven by pneumatic or magnetic pre-load ¹⁵ .	High-precision CNC machining of hardened tool steel or high-strength aluminum alloys ¹⁵ .	Concentrated point contacts induce high Hertzian stresses, causing rapid fatigue/creep in additive polymers; unsuited for large chassis footprints ⁴ .
EMERGE / Reconfigurable Modules [cite: 14]	Academic Modular Concept	Active mechanical finger grippers, passive alignment pins, and permanent magnet arrays ¹⁴ .	Desktop FDM 3D printing (PLA/ABS) with off-the-shelf servo actuators and fasteners ¹⁴ .	Extremely low structural load capacity (<5 kg); high joint play under dynamic load; low durability; TRL 3 research prototype ¹⁴ .
Generative Manipulators & UAV Frames [cite: 17, 18, 21]	Academic DfAM Research	Conventional cylindrical bolt circles integrated into defined preserve regions in	Industrial AM: DMLS Aluminum (AlSi10Mg), SLS Carbon-Filled PA12, FDM	Isolated component-level optimization (brackets/arms only); fails to optimize

		generative CAD ³ .	PEEK ⁴ .	overall mobile chassis; excludes quick-docking mechanisms ⁴ .
Topology-Optimized Chassis Brackets [cite: 19, 20]	Academic DfAM Research	Standard machined face-mounts bolted onto conventional rectangular steel tubing frames ¹⁹ .	Selective Laser Melting (SLM) or casting using topology-optimized CAD patterns ¹⁹ .	Fragmented optimization strategy; main structural chassis remains non-optimized subtractive tubing; zero functional module integration ⁴ .

Narrative Drawback Analysis Grouped by Theme

Interface Standardization and Tool-less Modularity

Commercial AMR designs treat functional top modules as semi-permanent aftermarket additions rather than integral elements of a unified system architecture². Mounting a top module to a commercial base platform requires technicians to align bolt patterns, secure threaded fasteners, remove protective body cowlings, and manually plug discrete wiring bundles into internal power distribution panels². This tool-bound installation process introduces operational downtime during facility reconfigurations.

Academic quick-change interfaces—such as kinematic couplings and pneumatic tool changers—provide tool-less modularity, but rely on concentrated point or line contact geometries to establish exact spatial constraint¹⁵. When subjected to dynamic warehouse forces, such as emergency braking shear or turning moment loads, these localized contact points generate high Hertzian stress concentrations¹⁶. When fabricated directly from additive polymers or unreinforced thermoplastics, these point-loading interfaces suffer localized yield, structural deformation, material creep, and premature fatigue failure⁴. Existing systems lack a tool-less mechanical coupling mechanism engineered to distribute structural load vectors across broad surface geometries optimized for additive manufacturing materials¹.

Structural Mass Efficiency and Payload-to-Weight Optimization

Commercial mobile robot frames are manufactured using subtractive cutting, sheet-metal

bending, plate welding, and structural steel box tubing². Because traditional manufacturing penalizes geometric complexity, commercial chassis designs rely on planar surfaces, right-angled corners, and uniform plate thicknesses⁴. This results in material placement in low-stress structural zones, creating a high chassis dead weight relative to actual payload capacity².

Carrying non-optimized chassis mass incurs operational penalties: it drains battery power, increases thermal loading on drive actuators, limits vehicle acceleration, and extends emergency stopping distances². While topological optimization algorithms can eliminate non-load-bearing mass, commercial manufacturers avoid these organic structures due to the high machining costs associated with multi-axis subtractive milling⁴.

Manufacturability and DfAM Misalignment

A divide exists between industrial additive manufacturing capabilities and structural chassis design practices:

Commercial mobile robot manufacturers avoid additive manufacturing for core structural frames, treating 3D printing as a rapid prototyping tool or relying exclusively on traditional sheet-metal weldments for primary structural assembly².

Academic generative design studies often produce unconstrained organic geometry that performs well in finite element simulations but presents physical manufacturing challenges during additive production³. Unconstrained generative studies routinely yield geometries with unprintable overhang angles that require dense internal support structures⁴. In hollow or enclosed chassis sections, these internal supports cannot be physically removed post-print⁴. Furthermore, unconstrained generative models frequently neglect additive process physics, such as thermal warping during Metal Powder Bed Fusion cooling or anisotropic inter-layer weakness along the Z-axis in material extrusion⁴. Crucially, pure generative algorithms treat chassis members as solid structural struts, failing to incorporate internal wire raceways, sensor mounting access, airflow cooling passages, or battery access bays¹.

Maintainability, Serviceability, and Cost Constraints

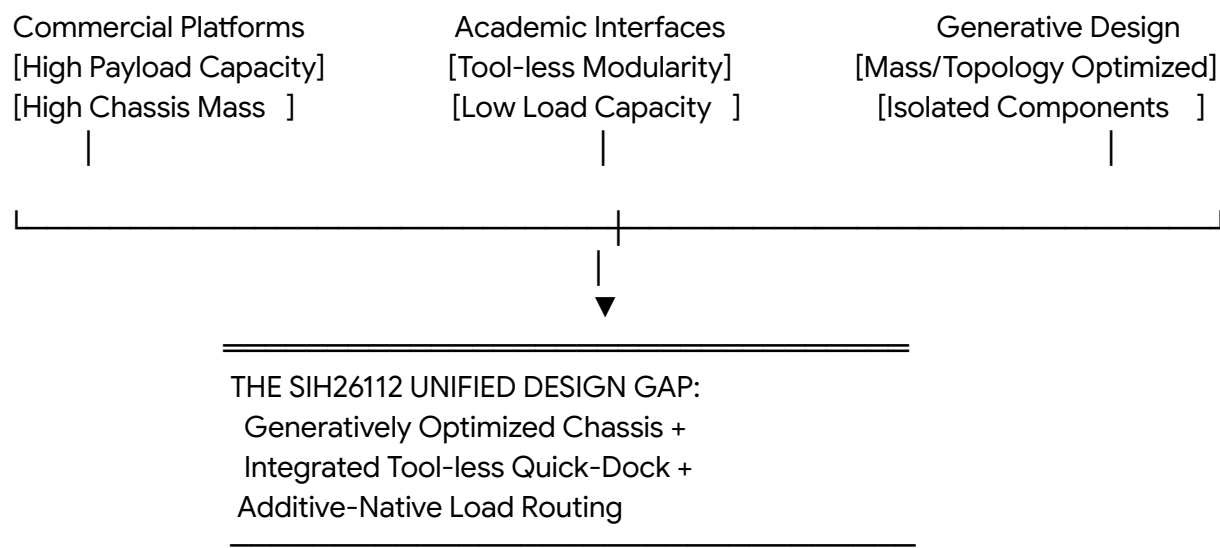
Commercial AMRs enclose their internal structural frames beneath non-structural aesthetic cowlings². Performing routine maintenance—such as inspecting drive belt tension, replacing a wheel motor controller, or swapping a depleted battery pack—requires removing multiple external panels and structural brackets². Furthermore, because traditional frames are fabricated as rigid, single-body weldments, structural damage caused by a warehouse collision requires replacing or cutting open the entire primary frame². Existing platforms lack modular, additively manufactured structural sub-assemblies designed for localized field replacement.

What is Genuinely Unsolved: The Targeted Technical Gap

Synthesizing the limitations across commercial, academic, and computational tiers reveals the

technical gap that Problem Statement SIH26112 targets¹. Current state-of-the-art solutions present an engineering trade-off: commercial platforms provide structural load capacity at the expense of high self-weight and tool-bound installation²; academic quick-release interfaces offer rapid modularity but are limited to lightweight end-effectors and suffer high localized contact stresses¹⁴; and generative design implementations produce lightweight structures but remain restricted to isolated sub-brackets¹⁷.

No existing system unifies an algorithmically optimized, additively manufactured structural chassis (Part A) with a tool-less, high-capacity modular attachment interface (Part B) into a single co-optimized mobile robotics platform¹.



Addressing this technical gap requires a design approach that satisfies three system-level engineering requirements simultaneously:

First, it requires System-Wide Structural Co-Optimization¹. Rather than treating the base chassis and the functional top attachment as separate mechanical parts connected by bolts, the structural load paths must be calculated as a single continuous system¹. Using Autodesk Fusion Generative Design, force vectors originating at the top attachment payload contact points must route directly through the modular interface locking nodes and down into the primary drive wheel suspension mountings¹.

Second, it requires an Additive-Native Quick-Docking Interface¹. The mechanical interface between Part A and Part B must eliminate tool-bound threaded fasteners while avoiding the concentrated point-loading characteristic of traditional kinematic couplings¹⁵. The interface must feature broad, organic contact geometries optimized for additive manufacturing materials (e.g., carbon-fiber-reinforced polymers or sintered metals), distributing dynamic acceleration, braking, and cornering shear forces evenly across the chassis frame¹.

Third, it requires Functional DfAM Integration¹. The generatively designed chassis members

must serve dual structural and operational roles¹. Internal load-bearing struts must be synthesized with hollow internal wire conduits, integrated sensor mounting housings, thermal dissipation fins, and self-supporting overhang geometries that eliminate unremovable internal print supports¹. Addressing these combined requirements defines the core technical challenge posed by the SIH26112 problem statement¹.

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